Scenario Analysis of a Delta-Hedged Option

(a) Option price. Write $\tau = T - t$. Under Q, the stock drifts are r, and the log returns are jointly normal. Their moments give

$$E^Q[(S_T^{(i)})^2 \mid S_t^{(i)} = s_i] = s_i^2 e^{(2r+\sigma_i^2)\tau}, \qquad E^Q[S_T^{(1)} S_T^{(2)} \mid S_t = s] = s_1 s_2 e^{(2r+\rho\sigma_1\sigma_2)\tau}.$$

Expanding the squared payoff and discounting therefore gives

$$\boxed{V(t, s_1, s_2; \sigma, \rho) = e^{r\tau}\left(s_1^2 e^{\sigma_1^2\tau} - 2s_1 s_2 e^{\rho\sigma_1\sigma_2\tau} + s_2^2 e^{\sigma_2^2\tau}\right).}$$

The deltas used for the initial hedge are

$$\partial_{s_1} V = 2e^{r\tau}\left(s_1 e^{\sigma_1^2\tau} - s_2 e^{\rho\sigma_1\sigma_2\tau}\right),$$
$$\partial_{s_2} V = 2e^{r\tau}\left(s_2 e^{\sigma_2^2\tau} - s_1 e^{\rho\sigma_1\sigma_2\tau}\right).$$

(b) Worst-case loss. I use the supplied parameters, with $r = 0.0132$, $T = 0.25$ and horizon $d = 5/252$ years. Initially,

$$V(0) = 921.739866, \qquad h_1 = -21.952543, \qquad h_2 = 31.201877.$$

The stock holdings are $(-h_1, -h_2)$, kept fixed for five days. Using the portfolio definition in the question,

$$\mathcal{V}_0 = V(0) - h_1 S_0^{(1)} - h_2 S_0^{(2)} = -921.739866,$$

$$\mathcal{V}_d = V(d, S_0^{(1)} e^{x_1}, S_0^{(2)} e^{x_2}; \sigma', \rho') - h_1 S_0^{(1)} e^{x_1} - h_2 S_0^{(2)} e^{x_2}, \qquad \ell = \mathcal{V}_0 - \mathcal{V}_d.$$

I enumerate all $6^4 \times 3 = 3{,}888$ scenarios and reprice with remaining maturity $T - d$. The largest loss is $\boxed{1{,}000.36}$.

Maximizing scenario	(b) Worst loss	(c) Worst weighted loss
Log returns (x_1, x_2)	$(0.20, 0.20)$	$(0.05, 0.05)$
Volatility factors $(\sigma_1'/\sigma_1, \sigma_2'/\sigma_2)$	$(0.5, 0.5)$	$(0.75, 0.75)$
New volatilities (σ_1', σ_2')	$(0.1107, 0.1518)$	$(0.16605, 0.2277)$
Correlation shift $\rho' - \rho$	$+0.5$	$+0.5$
New correlation ρ'	0.737	0.737
Portfolio loss ℓ	1,000.36	672.66

(c) Weighted scenario risk. I multiply the five factor weights from the question and maximize $w\ell$. The maximizing scenario is in the right column above. Its weight is

$$w = 1 \times 1 \times 1.25 \times 1.25 \times 1.25 = 1.953125, \qquad \max_n(w_n \ell_n) = \boxed{1{,}313.79}.$$

Its actual portfolio loss is $\boxed{672.66}$. The weights are used as given, without normalization. Since they can exceed one, the weighted risk can exceed the worst unweighted loss.

Python code and running instructions:
https://github.com/jesuswalking/MF-731-HW1/tree/main/HW4